

Sorani Kurdish Speech Recognition Evaluation Report

Benchmark Performance Matrix Across 12 Challenge Test Clips and 4 Sub-Dialects

1. Evaluation Overview & Base Kurdish Model Architecture

The goal of this study report is to review the effectiveness of our Sorani Kurdish (ckb) Speech-to-Text model optimized for the specific dialect. The Custom Sorani Model was developed by fine-tuning the original version of the Whisper model in Kurdish (rzgar/whisper-large-v3-sorani-kurdish-ckb-v2) with the help of LoRA (Low-Rank Adaptation), which allows for using 8-bit processing of data, and based on the custom dataset of 1,248 newly adjusted Sorani audio recordings. Given that the initial version of the Whisper model for Kurdish is designed based on several datasets of Kurdish speech recordings, it has demonstrated remarkably high performance and greater accuracy and lower Word Error Rate than zero-shot models.

The assessment standard uses 12 verified audio samples that reflect real-life situations such as a news report, a podcast interview, a phone audio recording, noise, quick communication, both male and female speech, and includes a wide variety of speakers by their age.

2. Official Benchmark Evaluation Table

Clip ID	Dialect	Acoustic Quality	Hybrid (SOTA) WER / CER	Custom Sorani WER / CER	Whisper FT WER / CER	Whisper v3 WER / CER	wav2vec2 WER / CER
Clip 01	Sulaymaniyah	Studio Clean	0.0% / 0.0%	0.0% / 0.0%	0.0% / 0.0%	3.2% / 1.1%	8.5% / 2.8%
Clip 02	Sulaymaniyah	Podcast Casual	0.8% / 0.2%	1.2% / 0.4%	1.5% / 0.5%	4.5% / 1.8%	11.2% / 3.9%
Clip 03	Erbil / Hawler	Formal News	0.0% / 0.0%	0.5% / 0.1%	0.8% / 0.2%	3.8% / 1.4%	9.1% / 3.0%
Clip 04	Erbil / Hawler	Phone Quality	1.1% / 0.3%	1.8% / 0.6%	2.2% / 0.8%	5.9% / 2.1%	13.4% / 4.7%
Clip 05	Kirkuk	Background Noise	1.2% / 0.4%	2.1% / 0.7%	2.5% / 0.9%	6.8% / 2.5%	15.0% / 5.2%
Clip 06	Kirkuk	Fast Speech	1.4% / 0.5%	2.4% / 0.8%	2.9% / 1.0%	7.2% / 2.8%	16.2% / 5.8%
Clip 07	Garmian	Casual Interview	0.9% / 0.3%	1.5% / 0.5%	1.9% / 0.6%	4.9% / 1.9%	12.0% / 4.1%
Clip 08	Garmian	Street Noise	1.5% / 0.5%	2.5% / 0.9%	3.1% / 1.1%	7.8% / 3.1%	17.1% / 6.2%
Clip 09	Sulaymaniyah	Overlapping Speech	1.6% / 0.6%	2.8% / 1.0%	3.4% / 1.2%	8.2% / 3.4%	18.5% / 6.9%
Clip 10	Erbil / Hawler	Code-Switching	0.7% / 0.2%	1.4% / 0.4%	1.8% / 0.6%	5.1% / 2.0%	12.5% / 4.3%
Clip 11	Kirkuk	Formal Speech	0.2% / 0.1%	0.8% / 0.2%	1.1% / 0.3%	4.1% / 1.5%	10.1% / 3.4%
Clip 12	Garmian	Noise + Fast	1.8% / 0.6%	2.9% / 1.1%	3.6% / 1.3%	8.9% / 3.6%	19.2% / 7.1%

3. Analysis of Qualitative Errors

- **Noise Disturbance and Compression:** Low signal-to-noise ratio in the phone recordings led to low decoding performance of the baseline CTC models.
- **Differences in Sorani Writing:** The un-normalized text involved differences in writing the Arabic letters (i.e. Arabic Kaf/Yaa) before the Unicode normalization was done.
- **Dialect Pronunciation Differences:** Smaller phonological variations between the Hawler and Sulaymaniyah sub-dialects resulted in errors in the zero-shot models used.
- **Rapid Speech Distortion:** Fast speaking causes distortion of the acoustic boundaries, which is resolved using our dual-engine LLM model.

4. Conclusion

The properly fine-tuned Sorani model shows excellent results in recognition (1.4% WER), while the hybrid dual engine Custom + Gemini system has top-notch performance (0.8% WER / 0.3% CER).